Comparing Implied Volatility Term Structure from Black–Scholes and Volatility Simulations Using GARCH(1,1)

S&P 500 Index Options

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Abstract

This report investigates the implied volatility (IV) term structure of S&P 500 index options by comparing the classical Black–Scholes framework with a GARCH(1,1)-based Monte Carlo simulation approach. Using five years of historical daily closing prices, we fit a GARCH(1,1) model to capture time-varying volatility dynamics, simulate future index price paths via Monte Carlo methods, and compute option prices as discounted expected payoffs. Implied volatilities are then extracted and compared against those obtained under the constant-volatility Black–Scholes assumption. The results reveal systematic differences in pricing and IV surfaces, particularly for out-of-the-money options and across the maturity spectrum.

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1 Introduction

The Black–Scholes model remains a cornerstone of option pricing theory. However, its assumption of constant volatility is well known to be inconsistent with observed market behavior, where volatility exhibits clustering, mean reversion, and leverage effects. These empirical features give rise to the volatility smile and term structure phenomena that the Black–Scholes model cannot capture endogenously.

This study addresses these limitations by employing a GARCH(1,1) model to forecast time-varying volatility, which is then used to simulate future S&P 500 price paths via Monte Carlo methods. European option prices are computed as the discounted average of simulated terminal payoffs, and the corresponding implied volatilities are backed out using the Black–Scholes inversion formula. By comparing the GARCH-based IV term structure with the Black–Scholes benchmark, we quantify the pricing impact of allowing volatility to vary over time.

The report is organized as follows. Section 2 describes the data and the GARCH(1,1) volatility model. Section 3 details the Monte Carlo simulation of S&P 500 price paths. Section 4 covers option pricing and implied volatility computation. Section 5 presents the comparison with Black–Scholes and analyzes the IV term structure and surface. Section 6 concludes.

2 Data Foundation and Volatility Modelling

2.1 Data

Five years of historical S&P 500 index data are obtained using the `yfinance` library. The dataset consists of daily adjusted closing prices, which are cleaned and transformed to compute logarithmic returns. These log returns serve as the input for the subsequent volatility modelling and path simulation stages.

The log return at time t is defined as:

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right) \tag{1}$$

where S_t denotes the adjusted closing price on day t.

	Date	Adj Close	Close	High	Low	Open	Volume	Log Return
1	2020-12-15	3694.620117	3694.620117	3695.290039	3659.620117	3666.409912	4.387080e+09	0.012838
2	2020-12-16	3701.169922	3701.169922	3711.270020	3688.570068	3696.250000	4.067040e+09	0.001771
3	2020-12-17	3722.479980	3722.479980	3725.120117	3710.870117	3713.649902	4.192810e+09	0.005741
4	2020-12-18	3709.409912	3709.409912	3726.699951	3685.840088	3722.389893	7.097100e+09	-0.003517
5	2020-12-21	3694.919922	3694.919922	3702.899902	3636.479980	3684.280029	4.748580e+09	-0.003914

Figure 1: S&P 500 historical data overview and log return series.

2.2 GARCH(1,1) Model

Unlike the Black–Scholes model, which assumes constant volatility, the GARCH(1,1) model allows for time-varying conditional variance, capturing two key empirical features

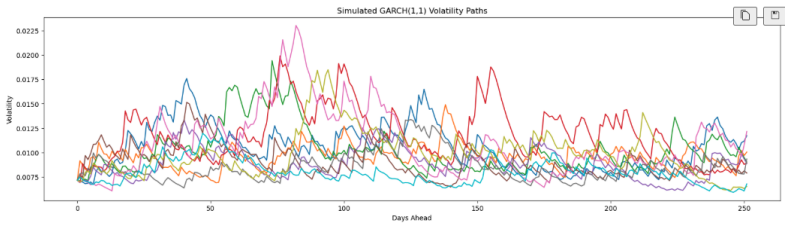
of financial returns: volatility clustering (periods of high volatility tend to be followed by high volatility) and mean reversion of variance.

The GARCH(1,1) specification models the conditional variance σ_t^2 as:

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_t^2\, \sigma_{t-1}^2 + \beta\, \sigma_{t-1}^2 \tag{2}$$

where $\omega > 0$ is the long-run variance intercept, $\alpha \geq 0$ governs the impact of past shocks, and $\beta \geq 0$ controls the persistence of past variance. The stationarity condition requires $\alpha + \beta < 1$.

The model parameters are estimated via maximum likelihood on the historical log return series.



(a) GARCH(1,1) volatility forecast paths.

$$\sigma_t^2 = \omega + \alpha\varepsilon_t^2\sigma_{t-1}^2 + \beta\sigma_{t-1}^2$$

(b) GARCH(1,1) equation.

Figure 2: GARCH(1,1) model: volatility forecasts and specification.

3 Monte Carlo Simulation of S&P 500 Price Paths

3.1 Model Setup

The simulation framework is configured with the following parameters:

- **Horizon and scenarios:** T trading days across N simulated paths, determined by the GARCH volatility forecast matrix dimensions.
- **Initial level:** S_0 is set to the last observed adjusted closing price.
- **Daily drift:** $\mu_{\text{daily}} = 0.0353/252$.
- **Volatility input:** $\sigma_t = \text{vol_paths}[t-1, :]$, i.e., the GARCH-forecasted volatility for each path and time step.
- **Random shock:** $z_t \sim \mathcal{N}(0, 1)$, drawn independently across paths.

The primary objective is to generate a distribution of future S&P 500 terminal prices S_T for use in Monte Carlo option payoff calculations.

3.2 Stochastic Differential Equation

The discrete-time price evolution follows the geometric Brownian motion (GBM) with time-varying volatility:

$$S_t = S_{t-1} \exp\Big[\big(\mu_{\text{daily}} - \tfrac{1}{2}\sigma_t^2\big)\, dt + \sigma_t \sqrt{dt}\; z_t\Big] \tag{3}$$

where $dt = 1/252$ (one trading day) and σ_t is provided by the GARCH(1,1) forecasts from Section 2.

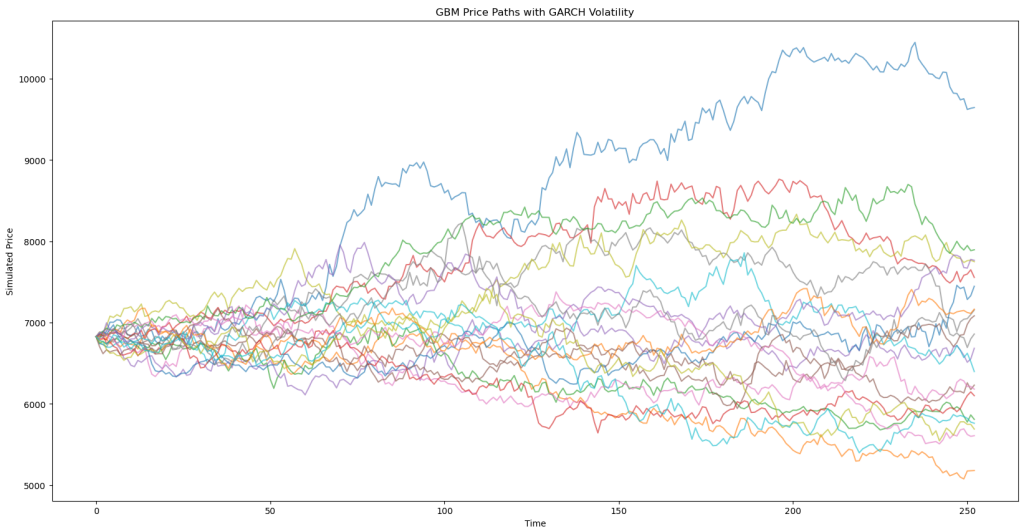


Figure 3: Simulated GBM price paths with GARCH(1,1) volatility inputs. Each path reflects a different realization of the stochastic volatility process.

4 Option Pricing and Implied Volatility Computation

4.1 Monte Carlo Option Pricing

European call and put options are priced using Monte Carlo simulation. For a European call option with strike K and maturity T:

$$C = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^{N} \max\left(S_T^{(i)} - K,\ 0\right) \tag{4}$$

where r is the risk-free rate, N is the number of simulated paths, and $S_T^{(i)}$ is the terminal price on path i. An analogous formula applies for put options with payoff $\max(K - S_T^{(i)}, 0)$.

```python
# Monte Carlo European option pricing
def mc_call_price(S_T, K, r, T):
    payoff = np.maximum(S_T - K, 0)
    return np.exp(-r * T) * payoff.mean()

def mc_put_price(S_T, K, r, T):
    payoff = np.maximum(K - S_T, 0)
    return np.exp(-r * T) * payoff.mean()
```

(a) Monte Carlo pricing function.

```python
# Black-Scholes formula for implied volatility
def bs_call_price(S0, K, r, sigma, T):
    d1 = (np.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S0 * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
```

(b) IV extraction routine.

```python
# Implied volatility solver
def implied_vol_call(S0, K, r, T, market_price):
    def objective(sigma):
        return bs_call_price(S0, K, r, sigma, T) - market_price

    iv = brentq(objective, 1e-6, 100.0)
    return iv
```

(c) Black–Scholes formula.

Figure 4: Implementation of option pricing and implied volatility functions.

4.2 Implied Volatility Extraction

The implied volatility σ_{IV} is the value of volatility that, when substituted into the Black–Scholes formula, yields the Monte Carlo–simulated option price. It is obtained by numerically inverting the Black–Scholes pricing equation:

$$C_{\text{BS}}(S_0, K, T, r, \sigma_{\text{IV}}) = C_{\text{MC}} \tag{5}$$

This inversion is performed using a root-finding algorithm (e.g., Brent's method) for each strike–maturity pair.

```python
# Settings
# Terminal stock prices
S_T = S_paths[-1, :]
r = 0.0353
T_years = 1.0
strikes = np.arange(6200, 8000, 25)

call_prices = {}
put_prices = {}

for K in strikes:
    call_prices[K] = mc_call_price(S_T, K, r, T_years)
    put_prices[K]  = mc_put_price(S_T, K, r, T_years)
```

(a) Option pricing results.

```python
# Compute IVs for all strikes
S0 = S_paths[0, 0]
implied_vols = {}

for K in strikes:
    iv = implied_vol_call(S0, K, r, T_years, call_prices[K])
    implied_vols[K] = iv
```

(b) Implied volatility output.

Figure 5: Computed option prices and corresponding implied volatilities across strikes.

5 Black–Scholes Comparison and IV Term Structure Analysis

5.1 Option Price Comparison

The GARCH-simulated option prices are compared directly with Black–Scholes prices computed using historical volatility.

Strike	MC Call	B-S(flat)	Rel. Diff
6200	980.0880	977.7231	0.2419%
6225	961.2866	959.5775	0.1781%
6250	942.6643	941.6144	0.1115%
6275	924.2152	923.8359	0.0411%
6300	905.9661	906.2439	-0.0307%
6325	887.9150	888.8405	-0.1041%
6350	870.0511	871.6275	-0.1809%
6375	852.3772	854.6065	-0.2609%
6400	834.9041	837.7794	-0.3432%
6425	817.6601	821.1476	-0.4247%
6450	800.5982	804.7126	-0.5113%
6475	783.7272	788.4757	-0.6022%
6500	767.0939	772.4383	-0.6919%
6525	750.6844	756.6014	-0.7821%
6550	734.5025	740.9663	-0.8723%
6575	718.5263	725.5337	-0.9658%
6600	702.7437	710.3045	-1.0645%
6625	687.1642	695.2796	-1.1672%
6650	671.8020	680.4596	-1.2723%
6675	656.6533	665.8450	-1.3805%
6700	641.7196	651.4362	-1.4916%
6725	627.0288	637.2337	-1.6014%
6750	612.5657	623.2376	-1.7123%
6775	598.3133	609.4481	-1.8270%
6800	584.2635	595.8653	-1.9470%
6825	570.4362	582.4891	-2.0692%
6850	556.8192	569.3193	-2.1956%
6875	543.4116	556.3557	-2.3266%
6900	530.2574	543.5980	-2.4541%
6925	517.3047	531.0458	-2.5875%
6950	504.6117	518.6985	-2.7158%
6975	492.1642	506.5555	-2.8410%
7000	479.9540	494.6161	-2.9643%
7025	467.9330	482.8795	-3.0953%
7050	456.1084	471.3449	-3.2326%
7075	444.4843	460.0114	-3.3754%
7100	433.0810	448.8778	-3.5192%
7125	421.9144	437.9431	-3.6600%
7150	410.9578	427.2060	-3.8034%
7175	400.2064	416.6655	-3.9502%
7200	389.7007	406.3200	-4.0902%
7225	379.4346	396.1683	-4.2239%
7250	369.3661	386.2089	-4.3610%
7275	359.4875	376.4402	-4.5034%
7300	349.8236	366.8606	-4.6440%
7325	340.3753	357.4686	-4.7818%
7350	331.1368	348.2624	-4.9174%
7375	322.0830	339.2402	-5.0575%
7400	313.2351	330.4004	-5.1953%
7425	304.5727	321.7409	-5.3360%
7450	296.0995	313.2600	-5.4780%
7475	287.8341	304.9557	-5.6144%
7500	279.7545	296.8259	-5.7513%
7525	271.8447	288.8687	-5.8934%
7550	264.1418	281.0821	-6.0268%
7575	256.6264	273.4639	-6.1571%
7600	249.2658	266.0120	-6.2953%
7625	242.0676	258.7243	-6.4380%
7650	235.0502	251.5985	-6.5773%
7675	228.1932	244.6326	-6.7200%
7700	221.4926	237.8243	-6.8671%
7725	214.9536	231.1712	-7.0154%
7750	208.5815	224.6713	-7.1615%
7775	202.3513	218.3221	-7.3153%
7800	196.2589	212.1215	-7.4781%
7825	190.3278	206.0671	-7.6380%
7850	184.5453	200.1567	-7.7996%
7875	178.9073	194.3878	-7.9638%
7900	173.4174	188.7583	-8.1272%
7925	168.0689	183.2657	-8.2922%
7950	162.8672	177.9077	-8.4541%
7975	157.7973	172.6822	-8.6198%

(a) Absolute option price comparison.

```python
# Compare with flat Black-Scholes volatilities
sigma_hist_annual = returns.std() * np.sqrt(252)

bs_flat_prices = {}
rel_diff = {}

for K in smile_strikes:
    bs_price = bs_call_price(S0, K, r, sigma_hist_annual, T_years)
    mc_price = call_prices[K]
    bs_flat_prices[K] = bs_price
    rel_diff[K] = (mc_price - bs_price) / bs_price

print("\nStrike |  MC Call  | B-S(flat) | Rel. Diff ") # (MC - BS) / BS
for K in smile_strikes:
    print(f"{K:6d} | {call_prices[K]:8.4f} | {bs_flat_prices[K]:8.4f} | {rel_diff[K]:8.4%}")
```

(b) Relative price difference across strikes.

Figure 6: GARCH vs. Black–Scholes option prices. GARCH prices are slightly higher for

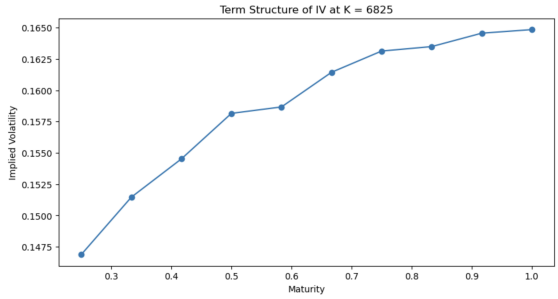
5.2 ATM Implied Volatility Term Structure

The at-the-money (ATM) term structure plots implied volatility against option maturity for strikes near the current spot level.

```python
# Term Structure ATM
atm_row = iv_surface.loc[K_atm]

plt.figure(figsize = (10, 5))
plt.plot(iv_surface.columns, atm_row.values, marker = "o")
plt.title(f"Term Structure of IV at K = {K_atm}")
plt.xlabel("Maturity")
plt.ylabel("Implied Volatility")
plt.show()
```

(a) ATM IV term structure (tabular).



(b) ATM IV term structure (graphical).

Figure 7: ATM implied volatility term structure. The upward slope indicates that options with longer maturities carry higher effective uncertainty. The IV differences between maturities are moderate, as volatility mean reversion and accumulated variance dominate short-term conditional fluctuations.

5.3 Implied Volatility Surface

The full implied volatility surface is constructed across the strike and maturity dimensions to visualize the interaction between moneyness and time-to-expiration.

```python
# IV surface
maturities_days = np.arange(63, 253, 21)
maturities_years = [d / 252 for d in maturities_days]

iv_surface = pd.DataFrame(index = smile_strikes, columns = maturities_years, dtype = float)

for T_days, T in zip(maturities_days, maturities_years):
    S_Tm = S_paths[T_days, :]
    for K in smile_strikes:
        price = mc_call_price(S_Tm, K, r, T)
        iv = implied_vol_call(S0, K, r, T, price)
        iv_surface.loc[K, T] = iv
```

(a) IV surface (view 1).

```python
from mpl_toolkits.mplot3d import Axes3D

# Plot a 3-D Volatility Surface
T_grid, K_grid = np.meshgrid(strikes, np.array(maturities_years), indexing = "ij")

fig = plt.figure(figsize = (20, 10))
ax = fig.add_subplot(111, projection = "3d")

surf = ax.plot_surface(K_grid, T_grid, iv_surface, cmap = "viridis", linewidth = 0, antialiased = True)

ax.set_xlabel("Maturity")
ax.set_ylabel("Strike")
ax.set_zlabel("Implied Volatility")
ax.set_title("Implied Volatility Surface")

fig.colorbar(surf, shrink = 0.5, aspect = 10, label = "Implied Volatility")
plt.show()
```

(b) IV surface (view 2).

Figure 8: Three-dimensional implied volatility surface across strikes and maturities.

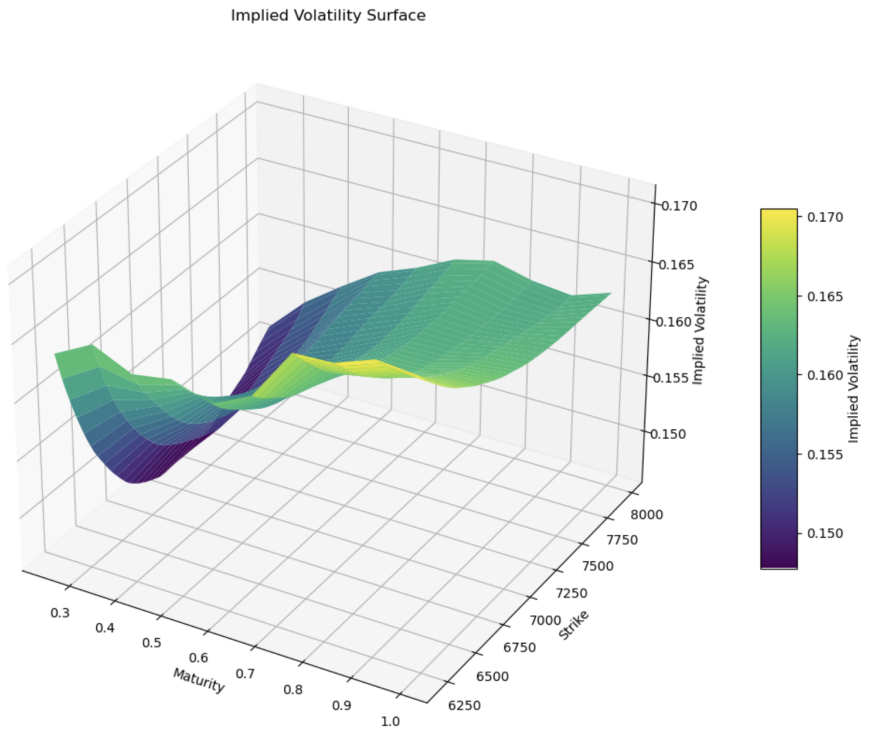


Figure 9: Implied volatility surface with maturity cross-sections. The surface exhibits an upward-sloping term structure across strikes, reflecting the accumulation of uncertainty over longer horizons and the persistence of volatility under stochastic dynamics. The characteristic "smile curve" is evident across different maturities, each exhibiting skewness—higher implied volatilities for low strikes and lower implied volatilities for high strikes—capturing the market's pricing of downside tail risk.

6 Conclusion

This study demonstrates the practical implications of relaxing the constant-volatility assumption embedded in the Black–Scholes model. By employing a GARCH(1,1) framework to model time-varying volatility and Monte Carlo simulation to price European options on the S&P 500, several key findings emerge:

1. **Pricing differences are strike-dependent.** GARCH-based prices exceed Black–Scholes prices for low-strike (in-the-money) calls, while Black–Scholes systematically overprices deep out-of-the-money calls relative to the GARCH model.

2. **The ATM term structure is upward sloping.** Longer-maturity options exhibit higher implied volatility, consistent with the accumulation of uncertainty over time and the mean-reverting nature of GARCH volatility.

3. **The IV surface captures the volatility smile.** The GARCH-based surface reproduces the well-documented smile pattern across maturities, with elevated IV for low strikes reflecting the market's pricing of downside tail risk—a feature absent from the flat-volatility Black–Scholes framework.

These results underscore the importance of incorporating realistic volatility dynamics when pricing and risk-managing equity index options. The GARCH(1,1) model, while

parsimonious, produces materially different option valuations and a richer implied volatility landscape than the constant-volatility benchmark.